

Pavan Seshu Kumar Poluparthi

✉ poluparthipavanseshukumar@gmail.com ☎ 9790564056 🌐 www.linkedin.com/in/pavan-seshu-kumar

SUMMARY

Full-Stack Developer with 7+ years of experience building scalable web applications and AI-powered platforms. React Developer with hands-on experience in creating AI-driven web interfaces, including chatbot UIs and seamless API integrations for intelligent applications. Specialized in React, Next.js, Node.js, and FastAPI, with a strong focus on responsive design, microservices-based architectures, and performance-driven solutions in collaborative environments.

SKILLS

Frontend: HTML, CSS, JavaScript (ES6+), Tailwind CSS, Material UI

Frontend Frameworks & Libraries: React.js, Next.js, Angular, React Native, Redux

Backend & APIs: Node.js, Express.js, Spring Boot, FastAPI, REST APIs

AI & Developer Tools: Cursor, Claude, Windsurf, Prompt Engineering, LLM API Integration, AI-assisted Development

Databases: MongoDB, MySQL

Development Practices: Agile Methodology, Responsive Design, Cross-Browser Compatibility

Tools & Platforms: Git, Postman

EXPERIENCE

Software Development Engineer II, Aspire Infolabs Global Pvt. Ltd

Aug 2018 – present

- **Developed and maintained web applications** using **React.js, React Native, and Node.js**, ensuring scalability, performance, and responsive design across devices.
- **Designed and implemented reusable React.js components** and shared libraries, improving consistency and maintainability across multiple modules.
- **Collaborated with Product Managers and UX Leads** to define project deliverables, refine requirements, and deliver seamless user experiences.
- **Created and integrated RESTful APIs**, handling data fetching, state management, and smooth front-end–back-end communication.
- **Implemented secure authentication flows** and data handling practices to ensure user privacy and data integrity.
- **Optimized application performance** through code refactoring, responsive layouts, and best practices in modern web development.
- Actively contributed to **debugging, troubleshooting, and enhancing existing codebases** to improve reliability and reduce technical debt.
- **Leveraged TypeScript** to write robust, type-safe, and maintainable code in both React and Node.js projects.

PROJECTS

ASSESS PRO – AI-Powered Online Assessment Platform (Next.js, React, Tailwind CSS, Shadcn UI, Fast API, REST APIs)

- Developed a **full-stack AI-powered online assessment platform** with separate learner and admin portals, enabling topic management, AI-generated question papers, proctored exams, and performance analytics.
- Built a responsive Next.js frontend using **reusable React components and Context API** for global state management (authentication, student data, notifications, and agent execution), ensuring scalable and maintainable architecture.
- Implemented end-to-end exam workflows supporting MCQ, coding, and descriptive questions, including features like **Monaco Editor integration (Python, SQL, Js, Java), exam timers, auto-save, and proctoring mechanisms** such as fullscreen enforcement and tab-switch detection.
- Designed conversational UI for an **AI chatbot interface** to assist users with assessment workflows, including requirement input, test strategy generation, and guided interactions, enhancing usability and automation.
- Developed an admin portal with **AI-assisted question paper generation**, agent simulation across skill levels (Beginner to Expert), and learner performance dashboards with custom data visualizations.

- Leveraged **AI-assisted development tools** to accelerate component generation, API integration, and debugging, reducing development time across complex asynchronous workflows.

AMC LADDER – Exam Preparation Platform (*Next.js, Tailwind CSS, Node.js, Razorpay*)

- Architected and launched a scalable **exam preparation platform for AMC CAT**, onboarding **1,000+ active users within the first 3 months**.
- Improved application performance by **~40%** through optimized **Next.js rendering strategies and Tailwind-based UI architecture**.
- Designed and implemented **Node.js backend services** for test management, analytics, and real-time progress tracking.
- Built detailed **performance reports and analytics dashboards** to help students track progress and identify improvement areas.
- Integrated **Razorpay payment gateway**, enabling secure subscription workflows and increasing successful transactions by **20%**.
- Adopted an **AI-assisted development workflow** to streamline debugging, generate boilerplate code, and accelerate implementation of new platform features.

HAPPILY HEALTH – Health & Wellness Mobile Platform (*React Native, Node.js*)

- Led development of a **cross-platform mobile application** with a team of **4 engineers**, achieving **10,000+ downloads in the initial release cycle**.
- Developed **scalable and reusable UI component architecture**, reducing feature development time by **30%**.
- Implemented **secure authentication, API integrations, and backend services**, ensuring compliance with healthcare data protection standards.
- Optimized application performance and user flows to improve engagement and usability.

CREDR – Used Bike & Scooter Marketplace (*React.js, Node.js, REST APIs*)

- Stabilized and enhanced a large-scale **React.js marketplace platform**, reducing production errors by **35%** and improving user retention.
- Developed an **internal dealer management tool** using React.js and Node.js that automated operational workflows, saving **~15 hours/week** for the operations team.
- Collaborated with **backend, product, and design teams** to deliver new platform features, contributing to a **15% increase in daily active users (DAUs)**.
- Improved API integrations and frontend performance to ensure a smoother user experience across devices.

MYCLNQ – Family Healthcare Application (*React Native, Node.js*)

- Developed key features including **appointment booking and clinic locator**, reducing average user wait time by **25%**.
- Designed **accessible and responsive UI components**, contributing to a **4.0★ rating on Google Play Store**.
- Integrated the mobile application with **Node.js backend services for appointment scheduling and medical records**, maintaining **99.9% uptime during peak usage**.
- Collaborated with backend teams to ensure reliable API communication and performance.

ACCELECOM SERVICES – Enterprise App Synchronization Platform (*Node.js, REST APIs*)

- Engineered **high-throughput synchronization services** between enterprise applications, processing **50,000+ daily sync events** with minimal latency.
- Optimized backend processing pipelines, reducing synchronization time from **3 seconds to under 1 second**.
- Designed and maintained **robust RESTful APIs**, enabling seamless integration with external client systems.
- Improved system reliability and reduced onboarding time for new integrations.

EDUCATION
